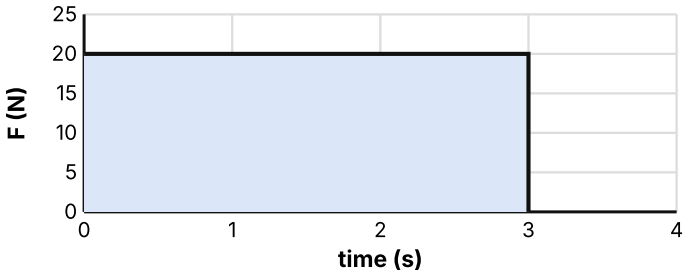
Mini-Lesson 4: Force–Time Graphs

AP Physics 1 · Unit 4 · Do with the mini-lesson before class Tue 11/17. Fill in each box when the lesson shows its orange tag.

Scan with your iPad camera to open the lesson

Name _____ Period _____

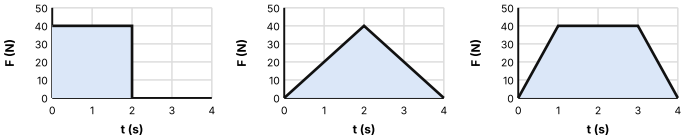
A The area is the impulse



Area = _____ × _____ = _____

The **area under an F–t graph** = _____ = _____. Area below the axis counts as _____.

B Any shape

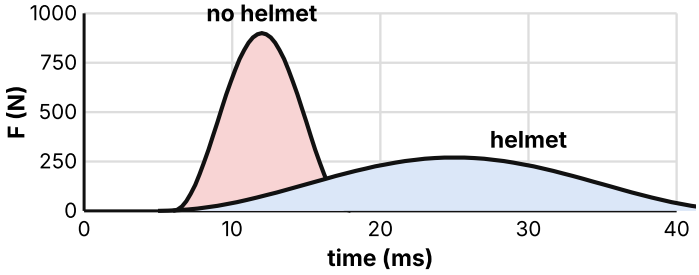


Rect: _____ Tri: _____ Trap: _____

Triangle area = _____. Trapezoid: split it into a _____.

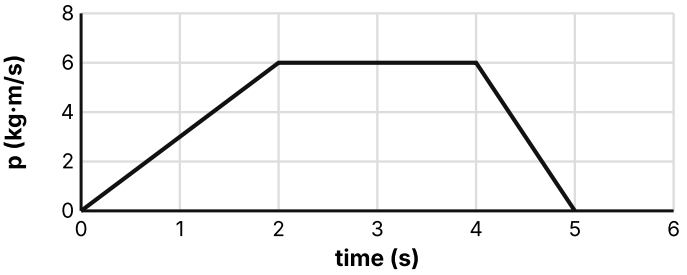
Same area means same _____, even with a different shape.

C Same area, different peak



Bigger impulse: _____. The helmet stretches out the _____, so the _____ is smaller.

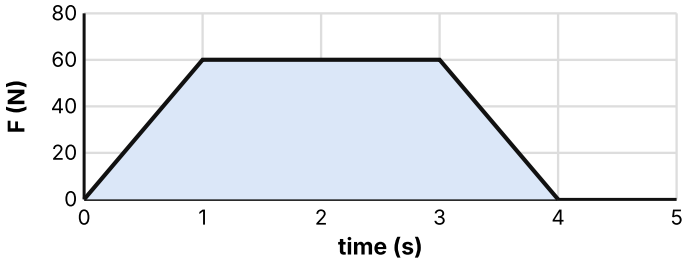
D Momentum–time graphs



Slope of p–t = _____. 0–2 s: _____ 2–4 s: _____ 4–5 s: _____

Flat p–t means zero _____, *not* zero momentum.

E Worked example: 12 kg cart, starts at rest

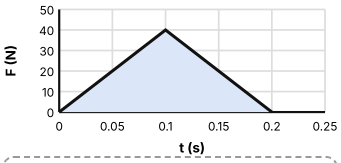


Areas: _____ + _____ + _____ = J = _____

v_f = J / m = _____

F Check questions: show your work

1. Impulse?



work:

J = _____

2. Net force?



work:

F = _____

3. 200 N for 0.010 s vs. 20 N for 0.10 s. More impulse: _____

Big idea in one sentence:

_____

_____